Tribhuvan University

Faculty of Science and Technology

OFFICE OF THE DEAN

Simulation and Modeling

CSIT — 5th Semester — 2082 — Answer Sheet

Subject Code: CSC328

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Section A

1. Describe the MARK and TABULATE block in GPSS. A coffee shop has a single barista serving customers. Customers arrive at an average rate of one every 3 minutes and the barista takes an average of 2.5 minutes to serve a customer. Create a simulation model and block diagram representing the coffee shop using GPSS blocks and simulate the system for 8 hours.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. What is simulation? Describe the analogy between a mechanical system and corresponding electrical system with reference to dynamic physical model.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. What are the properties of random number? The sequence of numbers 0.23, 0.45, 0.67, 0.12, 0.89, 0.34, 0.56, 0.78, 0.19, 0.41, 0.63, 0.08, 0.85, 0.29, 0.51, 0.73, 0.16, 0.94, 0.37, 0.59 has been generated. Test for the independence among numbers in the sequence starting with index (i) = 2 and lag (m)=3 using auto-correlation test. (α = 0.05, $Z\alpha/2$=1.96)

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Section B

Attempt any Eight questions[8×5=40]

4. Explain static mathematical model with suitable example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. Define Markov Chain. Explain with suitable example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. Explain non-stationary Poisson process in brief.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. Discuss the inverse transform technique and acceptance-rejection technique for random variate generation.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. Explain the iterative process of calibrating a model.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. Explain Kendall notation with appropriate example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. Explain Hybrid Simulation with example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. What is feedback system? Explain with suitable example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. Write short note on:a. Mid Square Methodb. Digital Analog Simulation

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]
